# Advanced Calculus

## Sequences in $\mathbb{R}$

Cauchy Sequences ・ Limits Supremum and Infimum

柯西數列 ・ 上極限與下極限

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§ 2.4

# Cauchy Sequences

柯西數列

# Recall: Tools from §2.3

先回顧上一節會用到的兩個工具

**Monotone Convergence Theorem (Thm 2.19)**

If $\{x_n\}$ is increasing and bounded above, or decreasing and bounded below, then $\{x_n\}$ converges to a finite limit.

**Bolzano–Weierstrass Theorem**

Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

- MCT needs **monotone**; BW only needs **bounded**.
- BW gives a convergent subsequence, not the whole sequence.

BW 只保證「有收斂的子數列」，不保證整個數列收斂，也不告訴你極限是多少。

# Definition: Cauchy Sequence

**Definition 2.27**

A sequence $\{x_n\}$ in $\mathbb{R}$ is **Cauchy** if for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that

$$n, m \geq N \implies |x_n - x_m| < \varepsilon.$$

**Convergent**

$$|x_n - a| < \varepsilon \quad (n \geq N)$$

Terms are close to a **fixed point** $a$.

**Cauchy**

$$|x_n - x_m| < \varepsilon \quad (n, m \geq N)$$

Terms are close to **each other**.

柯西數列：後面的項「彼此」越來越靠近。定義裡**沒有出現極限** $a$，這正是它好用的原因。

# Reading the Definition Carefully

## Order of quantifiers

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n, m \geq N : \ |x_n - x_m| < \varepsilon$$

$N$ may depend on $\varepsilon$, but **not** on $n$ or $m$.

## Negation: $\{x_n\}$ is NOT Cauchy

$$\exists \varepsilon_0 > 0 \ \forall N \in \mathbb{N} \ \exists n, m \geq N : \ |x_n - x_m| \geq \varepsilon_0$$

- Equivalent form: for $m > n$, write $m = n + p$ with $p \in \mathbb{N}$; Cauchy means $|x_{n+p} - x_n| < \varepsilon$ for **all** $p$ once $n \geq N$.

否定時「存在」與「任意」互換。證明「不是柯西」只要找到一個 $\varepsilon_0$，並在任意遠處找到一對 $n, m$。

# Main Results

## Remark 2.28

If $\{x_n\}$ is convergent, then $\{x_n\}$ is Cauchy.

## Theorem 2.29 (Cauchy)

Let $\{x_n\}$ be a sequence of real numbers. Then

$$\{x_n\} \text{ is Cauchy} \iff \{x_n\} \text{ converges to some } a \in \mathbb{R}.$$

**Structure of the argument** （證明架構）

$$\text{Cauchy} \implies \text{bounded} \overset{\text{BW}}{\implies} x_{n_k} \to a \implies x_n \to a$$

- The last step: $|x_n - a| \leq |x_n - x_{n_k}| + |x_{n_k} - a|$.

在 $\mathbb{R}$ 中「柯西 $\iff$ 收斂」，這依賴實數的完備性。

# Why Is This Useful?

## Convergence without knowing the limit

To prove $x_n \to a$ we need to know $a$. To prove $\{x_n\}$ is Cauchy we only compare terms of the sequence itself.

## Typical situations

- Recursively defined sequences.
- Partial sums $s_n = \sum_{k=1}^{n} a_k$ of a series (Chapter 6).
- Sequences given only by an inequality, e.g. $|x_{n+1} - x_n| \le c_n$.

## This fails in $\mathbb{Q}$

Choose $q_k \in \mathbb{Q}$ with $q_k \to \sqrt{2}$. Then $\{q_k\}$ is Cauchy, but has no limit in $\mathbb{Q}$.

「柯西 $\Rightarrow$ 收斂」不是理所當然的；在有理數裡就不成立。

# Technique 1: Telescoping + Geometric Tail

## Step 1: telescoping and triangle inequality

For $m > n$,

$$|x_m - x_n| = \left|\sum_{k=n}^{m-1}(x_{k+1} - x_k)\right| \le \sum_{k=n}^{m-1}|x_{k+1} - x_k|.$$

## Step 2: control the sum

If $|x_{k+1} - x_k| \le Cr^k$ with $0 < r < 1$, then

$$|x_m - x_n| \le C\sum_{k=n}^{m-1} r^k = Cr^n\,\frac{1 - r^{m-n}}{1 - r} < \frac{Cr^n}{1 - r}.$$

**Step 3:** the bound does **not depend on** $m$ and $\to 0$ as $n \to \infty$. Choose $N$ so that $\frac{Cr^N}{1-r} < \varepsilon$.

課本 Example 2.30 的手法。關鍵：找到只跟 $n$ 有關、且趨近 $0$ 的上界。

# Technique 2: Proving “Not Cauchy”

**Remark 2.31**

$x_{n+1} - x_n \to 0$ does **not** imply that $\{x_n\}$ is Cauchy.

**Textbook: $x_n = \log n$**

$$x_{n+1} - x_n = \log \tfrac{n+1}{n} \to 0$$

but $x_n \to \infty$, so not convergent, hence not Cauchy.

**Direct: $x_n = \sum_{k=1}^{n} \frac{1}{k}$**

$$x_{2n} - x_n = \sum_{k=n+1}^{2n} \frac{1}{k} \geq n \cdot \frac{1}{2n} = \frac{1}{2}.$$

Take $\varepsilon_0 = \frac{1}{2}$ and the pair $(2n, n)$.

**Two strategies:** (1) show it is unbounded / not convergent; (2) use the negation directly, often with $m = 2n$.

常用的一對是 $(n, 2n)$：差距一直維持在某個正數以上。

# Technique 3: Cauchy $\Rightarrow$ Bounded

**Every Cauchy sequence is bounded**

Take $\varepsilon = 1$: there is $N$ with $|x_m - x_N| < 1$ for $m \geq N$, so

$$|x_m| < 1 + |x_N| \quad (m \geq N), \qquad M := \max\{|x_1|, \ldots, |x_{N-1}|,\ 1 + |x_N|\}.$$

**When to use it**

- Products: add and subtract a middle term,

$$|x_n y_n - x_m y_m| \leq |x_n|\,|y_n - y_m| + |y_m|\,|x_n - x_m|,$$

  then bound $|x_n|, |y_m|$ by $M$.
- Whenever a problem says “without using Theorem 2.29”.

題目禁止用 Thm 2.29 時，「柯西 $\Rightarrow$ 有界」要自己用 $\varepsilon = 1$ 再證一次。

# Technique 4: From Series to Sequences

## Partial sums

For a series $\sum a_k$, let $s_n = \sum_{k=1}^{n} a_k$ and $s_0 = 0$. Then for $m \geq n \geq 1$,

$$s_m - s_{n-1} = \sum_{k=n}^{m} a_k.$$

So

$$\{s_n\} \text{ Cauchy} \iff \forall \varepsilon > 0 \, \exists N : \; m \geq n \geq N \Rightarrow \left| \sum_{k=n}^{m} a_k \right| < \varepsilon.$$

- This is the **Cauchy Criterion** for series (Theorem 6.8 later).
- For alternating sums, group terms in pairs to get upper / lower bounds.

把「級數收斂」轉成「部分和數列是柯西」，就只需要估計一段連續的和。

# § 2.4 Common Mistakes 常見錯誤

## Mistake 1: checking only consecutive terms

$|x_{n+1} - x_n| \to 0 \Rightarrow$ Cauchy  Counterexample: $x_n = \log n$.

## Mistake 2: letting $N$ depend on $m$

"Fix $m$; then $|x_n - x_m| < \varepsilon$ for $n$ large."  This only says each $x_m$ is "near" the tail; $N$ must work for **all** $n, m \geq N$.

## Mistake 3: circular use of Theorem 2.29

If the problem says "without Theorem 2.29", do **not** argue "Cauchy $\Rightarrow$ convergent $\Rightarrow$ bounded".

## Mistake 4: the bound depends on $m$

The final estimate must be $|x_m - x_n| <$ (something depending only on $n$).

# Think About It

動動腦

Is the sequence

$$x_n = \sum_{k=1}^{n} \frac{\sin k}{2^k}$$

convergent? (We cannot compute the limit explicitly.)

# Think About It: Answer

**Strategy:** we cannot guess the limit, so show $\{x_n\}$ is Cauchy (Technique 1).

For $m > n$,

$$
\begin{aligned}
|x_m - x_n| = \Big|\sum_{k=n+1}^{m} \frac{\sin k}{2^k}\Big| &\le \sum_{k=n+1}^{m} \frac{|\sin k|}{2^k} && \text{triangle inequality} \\
&\le \sum_{k=n+1}^{m} \frac{1}{2^k} && |\sin k| \le 1 \\
&= \frac{1}{2^n}\Big(1 - \frac{1}{2^{m-n}}\Big) < \frac{1}{2^n} && \text{geometric sum}
\end{aligned}
$$

Given $\varepsilon > 0$, choose $N$ with $\frac{1}{2^N} < \varepsilon$. Then $m > n \ge N \Rightarrow |x_m - x_n| < \varepsilon$.

**Answer**

$\{x_n\}$ is Cauchy, so by Theorem 2.29 it **converges**.

§ 2.5

# Limits Supremum and Infimum

上極限與下極限　（考試不考）

# Definition

## Definition 2.32

For a real sequence $\{x_n\}$,

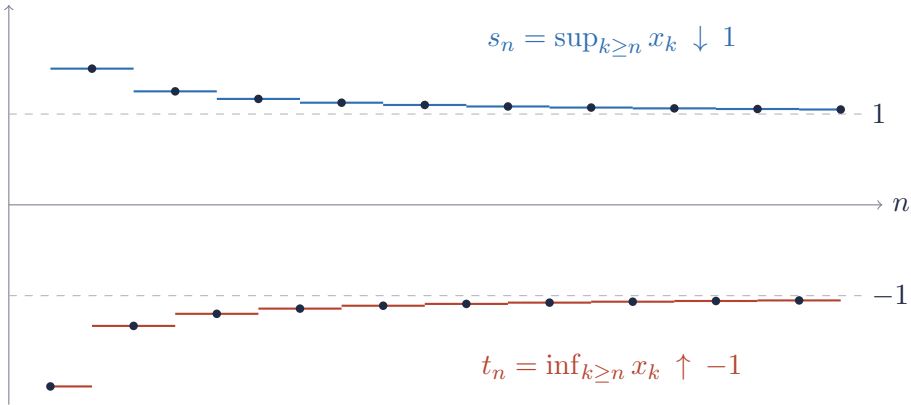
$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty}\left(\sup_{k\geq n} x_k\right), \qquad \liminf_{n\to\infty} x_n := \lim_{n\to\infty}\left(\inf_{k\geq n} x_k\right).$$

- $s_n := \sup_{k\geq n} x_k$ is **decreasing**, $t_n := \inf_{k\geq n} x_k$ is **increasing** (sup / inf over smaller and smaller sets).
- Hence both limits **always exist** in $[-\infty, \infty]$.
- Equivalent form:

$$\limsup_{n\to\infty} x_n = \inf_{n\in\mathbb{N}} \sup_{k\geq n} x_k, \qquad \liminf_{n\to\infty} x_n = \sup_{n\in\mathbb{N}} \inf_{k\geq n} x_k.$$

$\lim x_n$ 不一定存在，但 $\limsup$、$\liminf$ 一定存在（可能是 $\pm\infty$）。

## Picture: $x_n = (-1)^n\left(1 + \frac{1}{n}\right)$




$\limsup_{n\to\infty} x_n = 1$, $\liminf_{n\to\infty} x_n = -1$, $\lim_{n\to\infty} x_n$ does not exist.

藍線是「尾巴的上確界」，紅線是「尾巴的下確界」。

# Examples

## Example 2.33: $x_n = (-1)^n$

$\sup_{k\geq n}(-1)^k = 1$ and $\inf_{k\geq n}(-1)^k = -1$ for every $n$, so

$$\limsup_{n\to\infty} x_n = 1, \qquad \liminf_{n\to\infty} x_n = -1.$$

## Example 2.34: $x_n = 1 + \frac{1}{n}$

$\sup_{k\geq n} x_k = 1 + \frac{1}{n}$ and $\inf_{k\geq n} x_k = 1$, so

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = 1.$$

**How to compute:** write down $s_n$ and $t_n$ explicitly, then take limits; or find all subsequential limits (Theorem 2.37).

例 2.34 中 $\sup_n x_n = 2$，但 $\limsup x_n = 1$：上極限只看「尾巴」。

# Key Theorems

## Theorem 2.35

Let $s = \limsup_{n\to\infty} x_n$ and $t = \liminf_{n\to\infty} x_n$. There are subsequences $x_{n_k} \to s$ and $x_{\ell_j} \to t$.

## Theorem 2.36

For an extended real number $x$:

$x_n \to x \iff \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = x.$

## Theorem 2.37

$\limsup$ (resp. $\liminf$) is the **largest** (resp. **smallest**) value to which some subsequence converges: if $x_{n_k} \to x$, then

$$\liminf_{n\to\infty} x_n \le x \le \limsup_{n\to\infty} x_n.$$

$\limsup$ = 最大的子數列極限，$\liminf$ = 最小的子數列極限；兩者相等 $\iff$ 數列收斂。

# Properties

## Remarks 2.38, 2.39 and Theorem 2.40

- $\liminf_{n\to\infty} x_n \le \limsup_{n\to\infty} x_n$.
- $\{x_n\}$ bounded above $\iff \limsup_{n\to\infty} x_n < \infty$; bounded below $\iff \liminf_{n\to\infty} x_n > -\infty$.
- If $x_n \le y_n$ for $n$ large, then $\limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} y_n$ and $\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n$.

## Other useful facts

- $-\limsup_{n\to\infty} x_n = \liminf_{n\to\infty}(-x_n)$.
- $\limsup_{n\to\infty} x_n < r \;\Rightarrow\; x_n < r$ for all $n$ large.
- $\limsup_{n\to\infty} x_n > r \;\Rightarrow\; x_n > r$ for infinitely many $n$.
- $\limsup_{n\to\infty}(x_n + y_n) \le \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n$ (no $\infty - \infty$).

注意「$< r$」得到的是「最終都成立」，「$> r$」只得到「無窮多次成立」。

# § 2.5 Common Mistakes 常見錯誤

## Mistake 1: confusing $\limsup$ with $\sup$

$x_n = 1 + \frac{1}{n}$: $\quad \sup_n x_n = 2 \quad$ but $\quad \limsup_{n\to\infty} x_n = 1.$

## Mistake 2: thinking $\limsup$ may not exist

$\limsup$ and $\liminf$ **always exist** in $[-\infty, \infty]$; it is $\lim$ that may fail.

## Mistake 3: treating $\limsup$ as linear

$$\limsup_{n\to\infty}(x_n + y_n) = \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n$$

In general only "$\leq$" holds.

# One-Page Summary

| | |
|---|---|
| **Cauchy** | $\forall \varepsilon > 0\ \exists N:\ n, m \geq N \Rightarrow \lvert x_n - x_m \rvert < \varepsilon$ |
| **Thm 2.29** | In $\mathbb{R}$: Cauchy $\iff$ convergent |
| **Not Cauchy** | $\exists \varepsilon_0\ \forall N\ \exists n, m \geq N:\ \lvert x_n - x_m \rvert \geq \varepsilon_0$ |
| **Tools** | telescoping, geometric tail, Cauchy $\Rightarrow$ bounded, partial sums |
| $\limsup$ / $\liminf$ | $\lim\limits_{n\to\infty} \sup\limits_{k\geq n} x_k$, $\lim\limits_{n\to\infty} \inf\limits_{k\geq n} x_k$ (always exist) |
| **Thm 2.36** | $x_n \to x \iff \limsup x_n = \liminf x_n = x$ |
| **Thm 2.37** | largest / smallest subsequential limit |

Watch out: $x_{n+1} - x_n \to 0 \not\Rightarrow$ Cauchy $N$ must not depend on $m$ $\limsup \neq \sup$